



National Forensic Sciences University, Goa Campus Mid-semester Examination

Programme – M.Sc. DFIS		Sem – II	Date- 17.03.2026
Subject Name- Web Application Security		Subject Code- CTMSDFIS SII P2	
Time- 1.5 Hours		Max.Marks- 50	
Instructions: 1) Answer all questions. 2) Assume suitable data.			
Q.1	Solve any four	20 marks	
	a. Explain the role of different flags in the TCP header.	5 marks	
	b. Explain Transport Layer Security. How does it help the <i>HTTP</i> protocol?	5 marks	
	c. List out the different information received in the <i>response</i> method of the <i>HTTP</i> protocol.	5 marks	
	d. List out the recent 2025 OWASP top 10 attacks.	5 marks	
	e. Write a short note on <i>HTTP</i> fingerprinting	5 marks	
Q.2	Attempt all	15 marks	
	a. Explain the Vulnerability management lifecycle.	5 marks	
	b. List out the different consumers of CVE, and how they use it, explain with examples.	5 marks	
	c. Write a short note on a proxy server. How does it help the security team?	5 marks	
Q. 3	Attempt a and b	15 marks	
Q.3 a	Attempt any one		
Q.3 a	I. Explain the <i>Common Vulnerability Scoring System</i> in detail.	8 marks	
	OR		
	II. Explain different types of Web Authentication Mechanisms?	8 marks	
Q.3 b	Attempt any one	7 marks	
Q3 b	i. Explain the Secure Source Code Review process, why it is required, and under what circumstances you will prefer to go for automation.	7 marks	
	OR		
	II. Explain <i>STRIDE-based</i> threat modelling and how it differs from the <i>DREAD</i> model.	7 marks	